

Math 344, S19
Quiz 10

Name: Answer

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) True or false.

(a) The determinant of $A + B$ equals $\det A + \det B$.

False

$$A = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad A+B = \begin{bmatrix} 2 & 2 \end{bmatrix}$$
$$\det(A+B) = 4 \neq \det(A) + \det(B) = 1 + 1 = 2$$

(b) AB and BA have the same determinant.

True

$$\det(AB) = \det A \det B = \det B \det A = \det(BA)$$

(c) $\det \begin{bmatrix} a & d & 1 \\ b & e & 1 \\ c & f & 1 \end{bmatrix} = \det \begin{bmatrix} a & c & -b \\ d & f & -e \\ 1 & 1 & -1 \end{bmatrix}$

True

$$\begin{vmatrix} a & d & 1 \\ b & e & 1 \\ c & f & 1 \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ 1 & 1 & 1 \end{vmatrix} = - \begin{vmatrix} a & c & b \\ d & f & e \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} a & c & -b \\ d & f & -e \\ 1 & 1 & -1 \end{vmatrix}$$

(d) $\det(-A) = -\det(A)$.

False

$$A = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad \det(-A) = \begin{vmatrix} -1 & -1 \end{vmatrix} = 1$$
$$-\det A = - \begin{vmatrix} 1 & 1 \end{vmatrix} = -1$$

Problem 2. (6 points) Find the determinant of A using the co-factor expansion where

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & -3 \\ -2 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \det A &= \begin{vmatrix} \textcircled{1} & 0 & 0 & \textcircled{2} \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & -3 \\ -2 & 0 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 3 & 4 & 5 \\ 4 & 0 & -3 \\ 0 & 0 & \textcircled{1} \end{vmatrix} - 2 \cdot \begin{vmatrix} 0 & 3 & 4 \\ 5 & 4 & 0 \\ \textcircled{-2} & 0 & 0 \end{vmatrix} \\ &= 1 \cdot 1 \cdot \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix} - 2 \cdot (-2) \cdot \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix} \\ &= -16 + 4 \cdot (-16) \\ &= -80 \end{aligned}$$